

ORIE 6125 Final Project

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1 Overview

This report documents the implementation of first-order methods on the unconstrained optimization model

$$\min_{x \in \mathbb{R}^n} f(x) = \frac{1}{m} \sum_{i=1}^m \ell(\langle a_i, x \rangle - b_i) + \psi(x).$$

The delivered implementation contains the following components:

- a reproducible data generator with prescribed condition number for $A^\top A$;
- the gradient method (GD);
- the fast gradient method (FGM);
- the projected subgradient method for the ball-constrained formulation;
- experiment drivers that generate figures and summary tables.

2 Problem Formulation and Methodology

2.1 Data Generation

The assignment requires a random data generator for (A, b) such that

$$\sigma = \frac{\lambda_{\max}(A^\top A)}{\lambda_{\min}(A^\top A)}.$$

To enforce this directly, our implementation constructs

$$A = U \Sigma V^\top,$$

where $U \in \mathbb{R}^{m \times n}$ has orthonormal columns, $V \in \mathbb{R}^{n \times n}$ is orthogonal, and $\Sigma = \text{diag}(s_1, \dots, s_n)$ contains geometrically spaced singular values satisfying

$$\frac{s_1^2}{s_n^2} = \sigma.$$

Then

$$A^\top A = V \Sigma^2 V^\top,$$

so the eigenvalues of $A^\top A$ are exactly $s_1^2, \dots, s_n^2$, and therefore the condition number is controlled by this construction. The code uses a fixed random seed to make the data reproducible. The target and achieved condition numbers were numerically indistinguishable up to the displayed precision:

Target σ	Achieved $\kappa(A^\top A)$
10	10.00
10^3	1000.00
10^5	100000.00

Thus the data generator satisfies the requirement.

2.2 Smooth models

For the smooth models part of the assignment, the implementation solves the following optimization problem

$$\min_{x \in \mathbb{R}^n} \left[\frac{1}{m} \sum_{i=1}^m \ell(\langle a_i, x \rangle - b_i) + \frac{\mu}{2} \|x\|_2^2 \right],$$

with the following two loss functions:

$$\ell_{\text{quad}}(t) = \frac{1}{2}t^2, \quad \ell_{\text{log}}(t) = \log(1 + e^t).$$

Let $r = Ax - b$. Then the gradient under the quadratic loss is

$$\nabla f(x) = \frac{1}{m} A^\top r + \mu x$$

whereas under the logistic loss

$$\nabla f(x) = \frac{1}{m} A^\top \sigma(r) + \mu x, \quad \sigma(t) = \frac{1}{1 + e^{-t}}.$$

We use the following Lipschitz constants under the respective losses:

$$L_{\text{quad}} = \frac{\lambda_{\max}(A^\top A)}{m} + \mu, \quad L_{\text{log}} = \frac{\lambda_{\max}(A^\top A)}{4m} + \mu,$$

which follow respectively from the exact Hessian for the quadratic loss and the bound

$$\sigma(t)(1 - \sigma(t)) \leq 1/4$$

for the logistic loss (proven in Homework 1).

2.3 Projected subgradient model

For the nonsmooth part, the implementation solves the following optimization problem

$$\min_{\|x\|_2 \leq R} \frac{1}{m} \sum_{i=1}^m \ell(\langle a_i, x \rangle - b_i),$$

with three possible loss functions:

$$\ell_{\text{quad}}(t) = \frac{1}{2}t^2, \quad \ell_{\text{log}}(t) = \log(1 + e^t), \quad \ell_1(t) = |t|.$$

The (sub)gradient step update is

$$x_{k+1} = \Pi_{B_R(\mathbf{0})}(x_k - \gamma d_k),$$

where $d_k = g_k$ for the non-normalized version and

$$d_k = \frac{g_k}{\|g_k\|_2}$$

for the normalized version and Π is the projection operator. The Euclidean projection onto the ball is

$$\Pi_{B_R(\mathbf{0})}(y) = \begin{cases} y, & \|y\|_2 \leq R, \\ R \frac{y}{\|y\|_2}, & \|y\|_2 > R. \end{cases}$$

For the ℓ_1 loss function, the code uses the standard entry-wise sign subgradient.

2.4 Recorded histories and generated figures

Every method records the histories requested by the assignment:

- function values;
- gradient norms or subgradient norms;
- elapsed time snapshots;
- cumulative matrix-vector counts.

For each experiment case, the plotting routine saves the following figures:

- (i) residual $f(x_k) - f^\star$ versus iteration k ;
- (ii) gradient or subgradient norm versus iteration k ;
- (iii) residual versus elapsed time;
- (iv) residual versus cumulative matrix-vector products.

Thus the Python code produces the requested figures automatically.

3 Theoretical Expectations

Before discussing the computational results, it is useful to state the standard rates against which the experiments should be compared.

3.1 Smooth methods

For a convex differentiable objective with L -Lipschitz gradient, the gradient method has the following rate

$$f(x_k) - f^\star = O(1/k).$$

If the objective is additionally μ -strongly convex with $\mu > 0$, then GD becomes linearly convergent:

$$f(x_k) - f^\star \leq C \left(1 - \frac{\mu}{L}\right)^k.$$

For the fast gradient method (FGM), the expected smooth convex rate is faster:

$$f(x_k) - f^\star = O\left(\frac{1}{k^2}\right).$$

Under strong convexity, one expects accelerated linear convergence of the form

$$f(x_k) - f^* \leq C \left(1 - \sqrt{\frac{\mu}{L}}\right)^k.$$

Therefore, we expect that

- FGM should outperform GD in iteration complexity;
- increasing μ should improve convergence, because it strengthens convexity and improves effective conditioning;
- increasing σ should slow convergence, since it increases $\lambda_{\max}(A^\top A)$ and therefore the ratio between smoothness and strong convexity gets worse;
- adaptive search should not change the asymptotic order, but may improve practical convergence by selecting less conservative local step sizes.

3.2 Projected subgradient method

For general nonsmooth convex optimization, the projected subgradient method is expected to converge sublinearly. With theory-based step sizes tuned to the iteration budget, the canonical benchmark is

$$f(\bar{x}_k) - f^* = O\left(\frac{1}{\sqrt{k}}\right).$$

In practice, if one uses a fixed constant step size γ , then one often observes slower progress to a neighborhood whose size depends on γ , rather than clean asymptotic decay all the way to machine precision. Therefore, for the subgradient experiments, the most meaningful comparisons are qualitative: stability, monotonicity, sensitivity to step-size choice, and relative speed compared to the smooth methods.

4 Experimental Setup

The experiments were run with the moderate dimensions requested in the assignment:

$$n = 100, \quad m = 1000.$$

The data condition numbers were

$$\sigma \in \{10, 10^3, 10^5\}.$$

4.1 Smooth Methods Setup

For each pair (σ, loss) , where $\text{loss} \in \{\text{quadratic}, \text{logisitic}\}$, the report experiments compare:

- GD with fixed L ;
- FGM with fixed L ;
- GD with adaptive search;
- FGM with adaptive search.

The regularization parameter is varied over

$$\mu \in \{0, 10^{-3}, 10^{-1}\}.$$

The summary tables for smooth methods are based on 80 iterations. Representative figures shown in the report use 120 iterations so that the curve shapes are visually clearer.

For each configuration, the reference value f^* is estimated by the best objective value attained among the compared methods. This is permitted by the assignment and is sufficient here because the purpose is comparison of the algorithm trajectories on the same instance. When the residuals reach values around 10^{-14} to 10^{-16} , this should be interpreted as numerical saturation relative to the chosen proxy f^* , not as a literal proof of exact optimality.

4.2 Projected Subgradient Method Setup

For the projected subgradient method, the experiments use

$$R \in \{1, 5\}, \quad \gamma \in \{0.01, 0.05\},$$

with both normalized and non-normalized directions. The summary tables are based on 120 iterations, while representative figures use 180 iterations.

In the subgradient method, the reference value is again a best-observed proxy over the compared methods. Since the subgradient method is more sensitive to the fixed step size and the runs are finite, the absolute residual magnitudes should be read as empirical indicators rather than precise asymptotic certificates.

5 Results for the Smooth Methods

5.1 Aggregate comparison across σ

Table 1 reports the mean final residual after 80 iterations, averaged over the three values of μ for each (σ, loss) pair.

σ	loss	GD	FGM	GD-adapt	FGM-adapt
10	logistic	4.17e-09	2.83e-10	1.28e-13	1.15e-12
10	quadratic	2.90e-11	2.17e-11	3.04e-13	1.28e-13
10^3	logistic	4.98e-03	1.93e-04	2.10e-03	1.16e-13
10^3	quadratic	3.73e-03	1.16e-04	1.51e-03	2.78e-14
10^5	logistic	1.39e-02	1.83e-03	7.43e-03	1.00e-16
10^5	quadratic	9.08e-03	1.06e-03	4.39e-03	1.00e-16

Table 1: Average final residual after 80 iterations for the smooth methods, grouped by σ and loss and averaged over $\mu \in \{0, 10^{-3}, 10^{-1}\}$.

Several conclusions are immediate.

- (1) ***FGM is consistently better than GD in iteration count.***

At $\sigma = 10^3$ and 10^5 , the gap between FGM and GD is substantial for both losses, which is exactly what one expects from the $O(1/k^2)$ versus $O(1/k)$ smooth-convex theory and from accelerated linear convergence in the strongly convex regime.

(2) *Conditioning matters strongly.*

When $\sigma = 10$, all methods reach very small residuals within the allotted budget. As σ increases to 10^3 and 10^5 , the residuals of the fixed-step GD method become much larger, which matches the expected slowdown on ill-conditioned problems.

(3) *Adaptive search helps in practice.*

Both adaptive variants improve over their fixed-step counterparts in the harder regimes, and FGM with adaptive search is the strongest overall performer in these experiments.

(4) *The empirical behavior is consistent with theory.*

The tables do not directly prove asymptotic rates, but they strongly support the theoretical ranking predicted by first-order theory.

5.2 Effect of the strong convexity parameter μ

To isolate the role of strong convexity, Table 2 reports the final residuals for the hardest condition number, $\sigma = 10^5$, at each tested value of μ .

loss	μ	GD	FGM	GD-adapt	FGM-adapt
quadratic	0	1.06e-02	1.14e-03	5.49e-03	1.00e-16
quadratic	10^{-3}	1.10e-02	1.99e-03	5.84e-03	1.00e-16
quadratic	10^{-1}	5.67e-03	4.97e-05	1.83e-03	1.00e-16
logistic	0	1.96e-02	3.18e-03	1.13e-02	1.00e-16
logistic	10^{-3}	1.89e-02	2.31e-03	1.07e-02	1.00e-16
logistic	10^{-1}	3.10e-03	4.98e-07	3.20e-04	1.00e-16

Table 2: Final residuals after 80 iterations for the smooth methods at $\sigma = 10^5$.

The dominant trend is clear: larger μ improves convergence, especially for the fixed-step methods. This agrees with the theory because larger μ strengthens convexity and decreases the effective condition number $\frac{L}{\mu}$ whenever $\mu > 0$. The improvement is particularly visible for the logistic loss, where the residual for fixed-step FGM drops from roughly 3.18×10^{-3} at $\mu = 0$ to about 4.98×10^{-7} at $\mu = 10^{-1}$. The same qualitative improvement appears for GD and GD-adaptive.

A minor nuance is that the improvement from $\mu = 0$ to $\mu = 10^{-3}$ is not always dramatic within only 80 iterations, especially when σ is very large. This is not a contradiction to theory: a tiny value of μ only mildly changes the condition number, so finite-time improvement can remain modest. The larger value $\mu = 10^{-1}$ produces the clearest benefit.

5.3 Representative figures and rate interpretation

Figure 1 shows a representative smooth-convex case: quadratic loss, $\sigma = 10^3$, and $\mu = 0$.

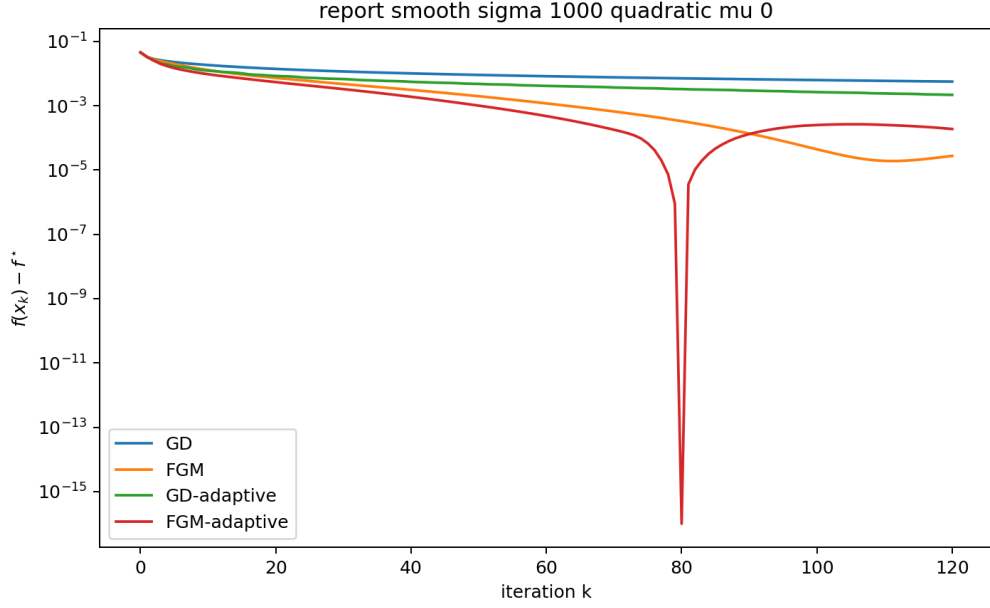


Figure 1: Residual versus iteration for quadratic loss with $\sigma = 10^3$ and $\mu = 0$.

This figure is consistent with the expected ordering in the merely convex regime: FGM and especially FGM-adaptive descend more rapidly than GD. Since $\mu = 0$, one should not expect linear convergence in the strict sense; the main theoretical comparison here is the faster sublinear decrease of FGM relative to GD, which is exactly what the figure shows.

Figure 2 shows logistic loss with moderate conditioning and substantial regularization.

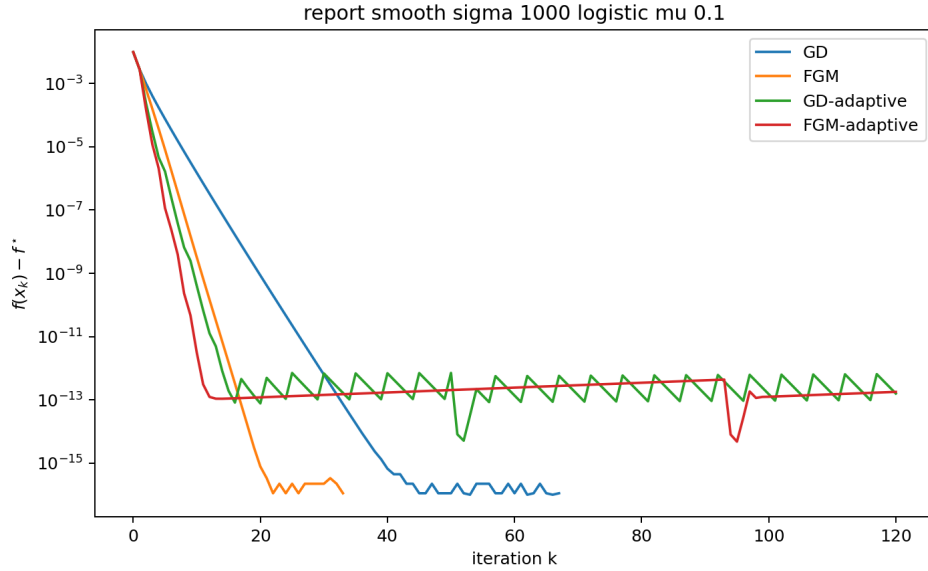


Figure 2: Residual versus iteration for logistic loss with $\sigma = 10^3$ and $\mu = 10^{-1}$.

Here the curves bend downward much more sharply and the best methods reach the numerical floor of the chosen proxy. This behavior is exactly what one expects once strong convexity is present: the decay is much closer to a straight line on a semi-logarithmic plot, which is the practical signature of linear convergence.

Finally, Figure 3 shows a difficult strongly convex instance with very large condition number.

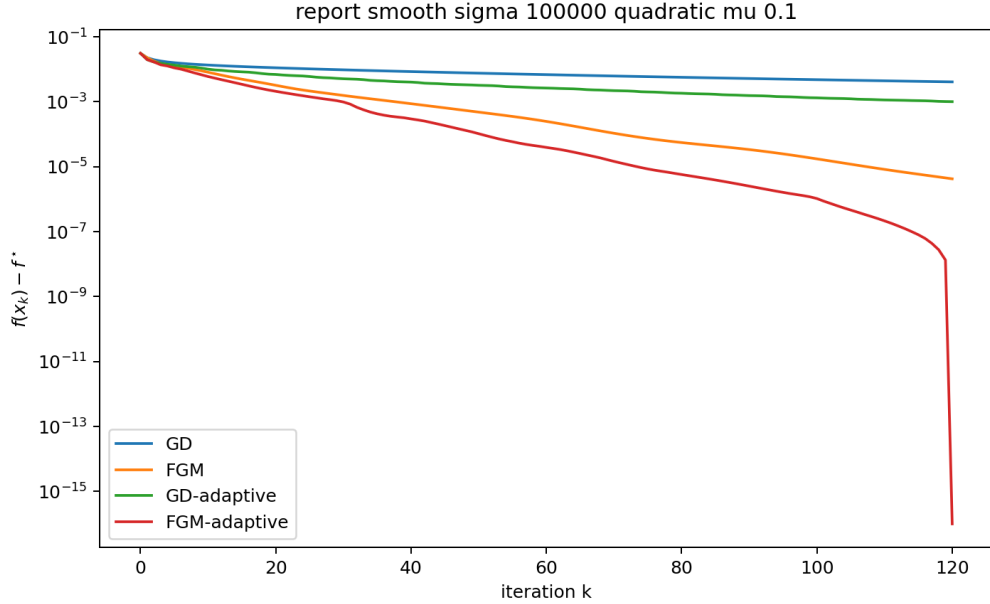


Figure 3: Residual versus iteration for quadratic loss with $\sigma = 10^5$ and $\mu = 10^{-1}$.

Even under severe ill-conditioning, acceleration remains highly beneficial, and adaptive search remains helpful. Thus the experiments support the following expectations:

- GD behaves like a first-order method with the standard smooth-convex / strongly-convex rates;
- FGM is faster than GD and especially valuable on difficult instances;
- increasing μ improves convergence;
- increasing σ slows convergence;
- adaptive search improves practical speed but does not change the theoretical order.

5.4 Does adaptive search help?

Yes!!! The strongest evidence is that FGM-adaptive method repeatedly reaches the smallest residuals in the ill-conditioned cases, often all the way down to the numerical floor of the chosen f^* proxy. The GD-adaptive method also improves over fixed-step GD at $\sigma = 10^3$ and 10^5 .

However, one should be careful in their conclusion. Adaptive search does not create a new asymptotic convergence class. Rather, it improves the practical step sizes by avoiding a single conservative global estimate. Therefore, the correct interpretation is that "Adaptive search helps the methods converge faster in finite time on these instances, but its role is practical rather than asymptotic".

6 Results for the Projected Subgradient Method

6.1 Aggregate comparison

Table 3 reports the average final residual after 120 iterations, averaging over $R \in \{1, 5\}$ and $\gamma \in \{0.01, 0.05\}$.

σ	loss	subgrad	normalized
10	ℓ_1	2.99e-02	3.81e-07
10	logistic	1.72e-02	1.00e-16
10	quadratic	2.35e-02	1.00e-16
10^3	ℓ_1	2.34e-02	9.74e-05
10^3	logistic	3.20e-02	1.00e-16
10^3	quadratic	2.05e-02	1.00e-16
10^5	ℓ_1	4.00e-01	9.18e-06
10^5	logistic	8.75e-03	9.17e-04
10^5	quadratic	3.25e+02	1.00e-16

Table 3: Average final residual after 120 iterations for the projected subgradient method.

The main practical conclusion is that normalization makes the method much more robust. On difficult instances, the non-normalized method becomes highly sensitive to the step size, whereas the normalized method is consistently stable. The most extreme example is the quadratic-loss case with $\sigma = 10^5$, where the average non-normalized residual becomes very large.

6.2 Detailed behavior in the hardest regime

Table 4 shows all tested (R, γ) combinations for $\sigma = 10^5$.

loss	R	γ	subgrad	normalized
quadratic	1.0	0.01	1.93e-03	1.00e-16
quadratic	1.0	0.05	5.00e+01	1.00e-16
quadratic	5.0	0.01	1.93e-03	1.00e-16
quadratic	5.0	0.05	1.25e+03	1.00e-16
logistic	1.0	0.01	1.75e-02	1.00e-16
logistic	1.0	0.05	1.00e-16	1.83e-03
logistic	5.0	0.01	1.75e-02	1.00e-16
logistic	5.0	0.05	1.00e-16	1.83e-03
ℓ_1	1.0	0.01	1.00e-03	1.84e-05
ℓ_1	1.0	0.05	7.99e-01	1.00e-16
ℓ_1	5.0	0.01	1.00e-03	1.84e-05
ℓ_1	5.0	0.05	7.99e-01	1.00e-16

Table 4: Projected subgradient results at $\sigma = 10^5$ after 120 iterations.

These numbers should be interpreted with some care.

- (1) The method is being run for a fixed finite budget with fixed step sizes, not with an iteration-budget-tuned schedule. Therefore one should not expect a perfect textbook $O\left(\frac{1}{\sqrt{k}}\right)$ signature in every finite run.
- (2) The reference value is a best-observed proxy, so a residual of order 10^{-16} means that a run matched the best result observed among the compared methods, not that the global optimum was analytically certified.
- (3) The very large residuals in the non-normalized quadratic case are not a contradiction to theory; they reflect the instability of using a fixed step size with large subgradient norms on ill-conditioned instances.

With those caveats stated explicitly, the qualitative conclusion is nevertheless fully meaningful: the subgradient method is slower and much more step-size-sensitive than the smooth methods, and normalization can dramatically improve robustness.

6.3 Representative figures

Figure 4 shows the ℓ_1 -loss case with moderate ill-conditioning.

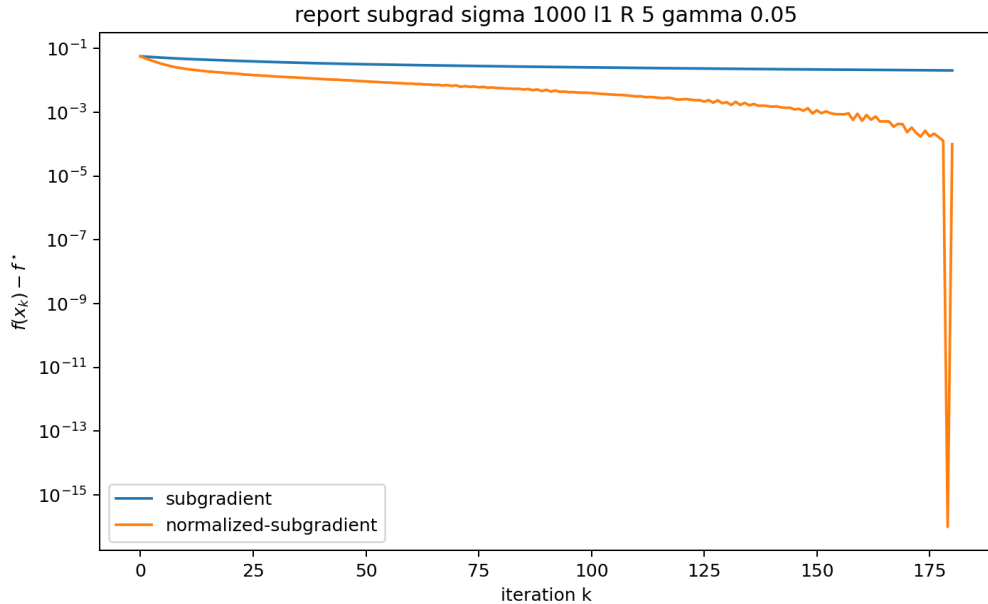


Figure 4: Residual versus iteration for the projected subgradient method on the ℓ_1 -loss problem with $\sigma = 10^3$, $R = 5$, and $\gamma = 0.05$.

The descent is much slower than in the smooth GD/FGM experiments, which agrees with the standard sublinear theory. The normalized method is also visibly more regular.

Figure 5 shows logistic loss with good conditioning.

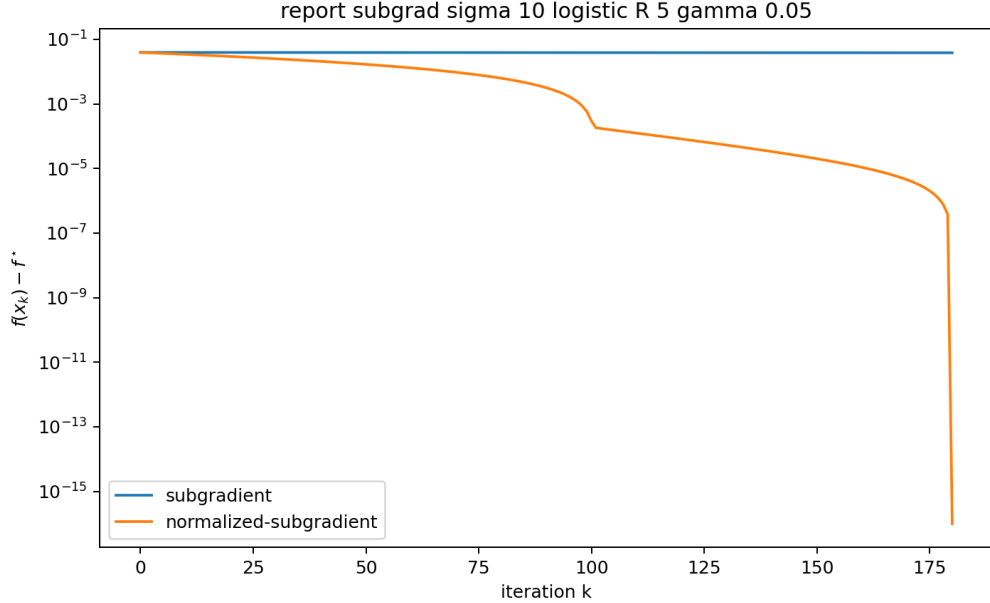


Figure 5: Residual versus iteration for the projected subgradient method on logistic loss with $\sigma = 10$, $R = 5$, and $\gamma = 0.05$.

Even here, the method is not competitive with the smooth first-order methods in terms of residual decay over the same number of iterations. This is exactly what theory suggests: the subgradient method should be regarded as the most general but also the slowest method in the current assignment.

7 Concluding Remarks

We report the following remarks based on the experiments.

What is the rate of convergence for each method? For the smooth problems, GD behaves consistently with the standard $O\left(\frac{1}{k}\right)$ rate in the merely convex regime and linear convergence in the strongly convex regime. FGM behaves consistently with the improved $O\left(\frac{1}{k^2}\right)$ smooth-convex rate and accelerated linear convergence when $\mu > 0$. The projected subgradient method is noticeably slower and behaves qualitatively like a subgradient scheme with sublinear convergence.

How is the rate affected by μ ? Increasing μ improves convergence. This is visible in the tables and figures, especially for the most ill-conditioned instances. The effect is strongest when moving to the larger regularization level $\mu = 10^{-1}$, where the methods exhibit much faster decay.

How is the rate affected by σ ? Increasing σ slows convergence. This effect is clear in the fixed-step smooth methods and also in the sensitivity of the subgradient method. Large condition numbers create more difficult instances and widen the practical gap between basic and accelerated methods.

Does adaptive search help? Yes, adaptive search helps in practice. It does not change the theoretical convergence guarantee, but it improves finite-time performance by avoiding overly conservative fixed step sizes. The best overall empirical method in these experiments is adaptive FGM.

Do the results agree with theory? Yes, overall they do. The ranking of methods, the beneficial effect of strong convexity, and the harmful effect of ill-conditioning all match the theory. The only place where care is needed is the projected subgradient method: Since the experiments use finite runs, fixed step sizes, and a best-observed proxy for f^* , the interpretation there should remain qualitative rather than over-quantified. Once that is ensured, the experimental findings still align well with theoretical expectations.